

МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ РОССИЙСКОЙ ФЕДЕРАЦИИ

Федеральное государственное автономное образовательное учреждение высшего образования
Санкт-Петербургский национальный исследовательский университет ИТМО
Мегафакультет трансляционных информационных технологий

Факультет информационных технологий и программирования

Лабораторная работа №3

По дисциплине «Проектирование и реализация баз данных»
Вариант №19

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Санкт-Петербург

Цель работы: приобретение практических навыков проектирования и реализации базы данных в СУБД PostgreSQL с использованием pgAdmin 4, создания таблиц и ограничений целостности, заполнения базы данных рабочими данными, выполнения резервного копирования и восстановления базы данных.

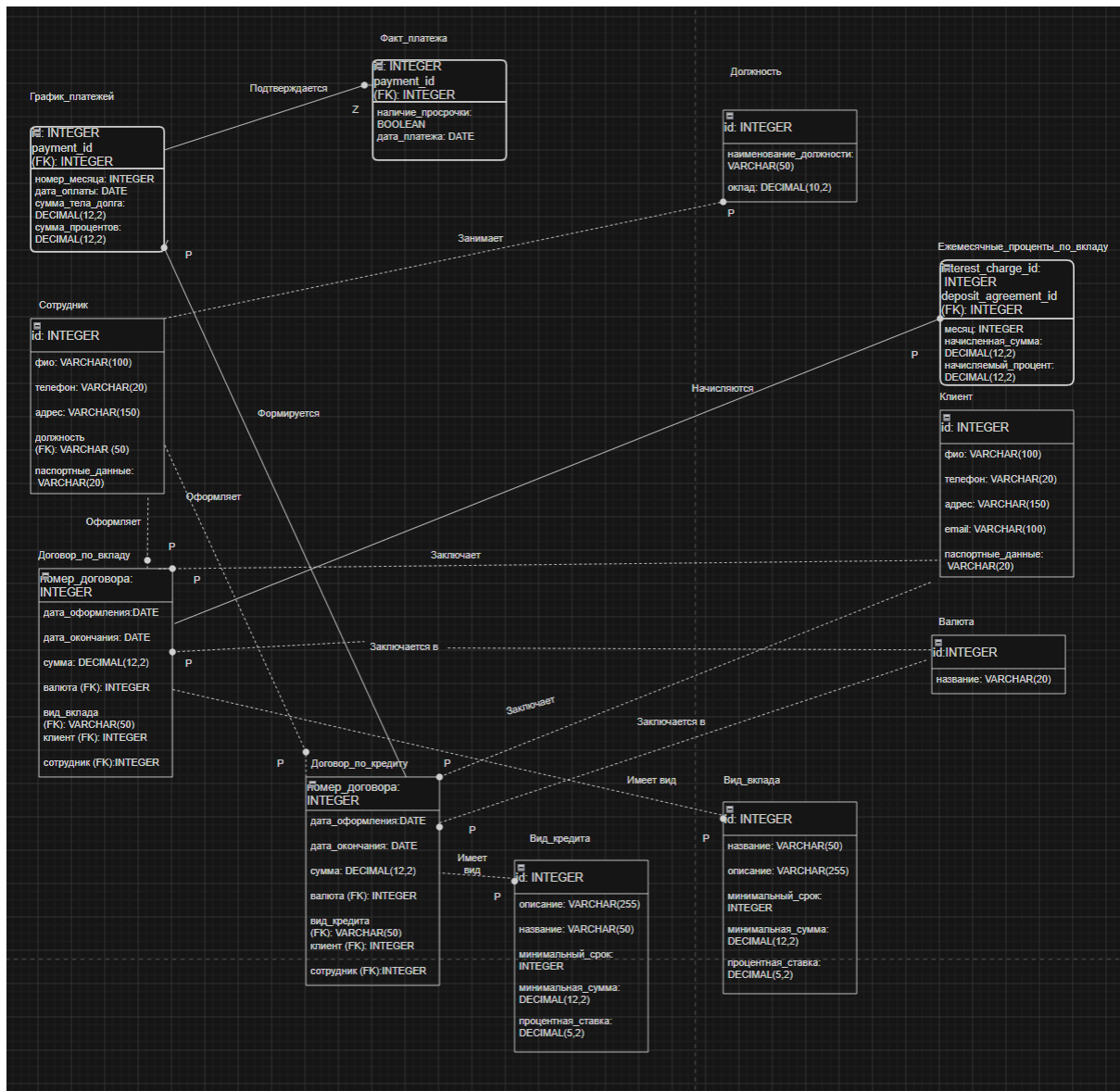
Практическое задание:

1. Создать базу данных с использованием pgAdmin 4.
2. Создать схему в составе базы данных.
3. Создать таблицы базы данных в соответствии с разработанной моделью предметной области.
4. Установить ограничения на данные: Primary Key, Foreign Key, Unique, Check.
5. Заполнить таблицы базы данных рабочими данными.
6. Создать резервную копию базы данных в форматах Plain и Custom.
7. Выполнить восстановление базы данных из резервной копии.

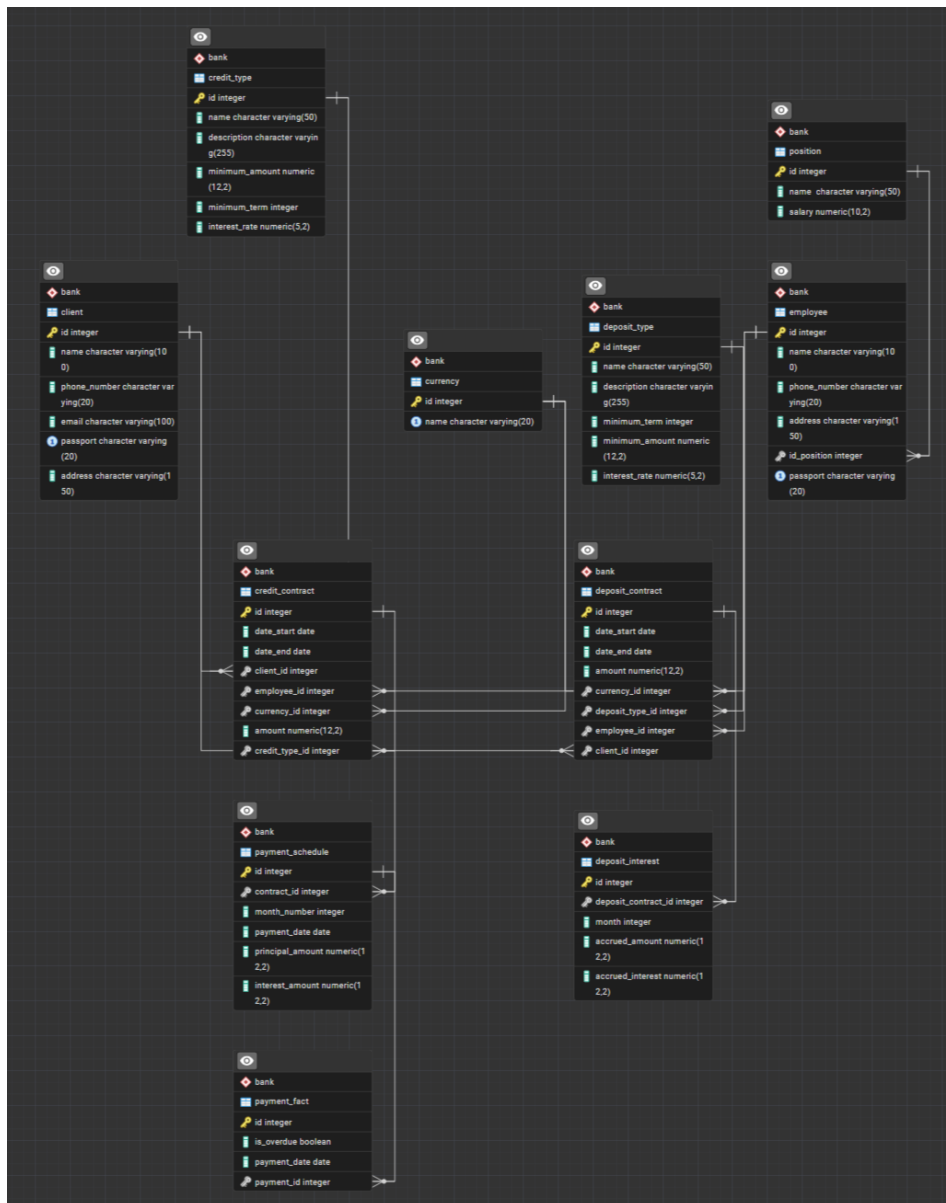
Выполнение:

I Название: fpi_bd, вариант 19

II



<https://drive.google.com/file/d/1ug-9XubDbyQjM2TLJ5fzXY6wW8d81G2J/view?usp=sharing>



IV

```
--
-- PostgreSQL database dump
--

\restrict zHfzVYNHwb6rxDaU6YUMbY1q8VzHSjoQFRuHLBvpvEsoPPyVqJhaAQKvY8jHipL

-- Dumped from database version 18.3
-- Dumped by pg_dump version 18.3

-- Started on 2026-04-05 18:35:52

SET statement_timeout = 0;
SET lock_timeout = 0;
```

```

SET idle_in_transaction_session_timeout = 0;
SET transaction_timeout = 0;
SET client_encoding = 'UTF8';
SET standard_conforming_strings = on;
SELECT pg_catalog.set_config('search_path', '', false);
SET check_function_bodies = false;
SET xmloption = content;
SET client_min_messages = warning;
SET row_security = off;

--
-- TOC entry 6 (class 2615 OID 16389)
-- Name: bank; Type: SCHEMA; Schema: -; Owner: postgres
--

CREATE SCHEMA bank;

ALTER SCHEMA bank OWNER TO postgres;

SET default_tablespace = '';

SET default_table_access_method = heap;

--
-- TOC entry 222 (class 1259 OID 16408)
-- Name: client; Type: TABLE; Schema: bank; Owner: postgres
--

CREATE TABLE bank.client (
    id integer NOT NULL,
    name character varying(100) NOT NULL,
    phone_number character varying(20),
    email character varying(100),
    passport character varying(11) NOT NULL,
    address character varying(150),
    CONSTRAINT client_id_positive CHECK ((id > 0)),
    CONSTRAINT client_phone_check CHECK (phone_number ~ '^\\+?[0-9]{11}$'),
    CONSTRAINT client_email_check CHECK (email ~ '^\\[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\\.\\[A-Za-z]{2,}$'),
    CONSTRAINT client_passport_format_check CHECK (passport ~ '^\\[0-9]{4}\\s?[0-9]{6}$')
);

```

```
ALTER TABLE bank.client OWNER TO postgres;
```

```
--  
-- TOC entry 227 (class 1259 OID 16496)  
-- Name: credit_contract; Type: TABLE; Schema: bank; Owner: postgres  
--
```

```
CREATE TABLE bank.credit_contract (  
    id integer NOT NULL,  
    date_start date NOT NULL,  
    date_end date NOT NULL,  
    client_id integer NOT NULL,  
    employee_id integer NOT NULL,  
    currency_id integer NOT NULL,  
    amount numeric(12,2) NOT NULL,  
    credit_type_id integer NOT NULL,  
    CONSTRAINT positive_credit_amount CHECK ((amount > (0)::numeric)),  
    CONSTRAINT valid_deposit_dates CHECK ((date_end > date_start))  
);
```

```
ALTER TABLE bank.credit_contract OWNER TO postgres;
```

```
--  
-- TOC entry 225 (class 1259 OID 16448)  
-- Name: credit_type; Type: TABLE; Schema: bank; Owner: postgres  
--
```

```
CREATE TABLE bank.credit_type (  
    id integer NOT NULL,  
    name character varying(50) NOT NULL,  
    description character varying(255),  
    minimum_amount numeric(12,2) NOT NULL,  
    minimum_term integer NOT NULL,  
    interest_rate numeric(5,2) NOT NULL,  
    CONSTRAINT positive_interest_rate CHECK ((interest_rate >= (0)::numeric)),  
    CONSTRAINT positive_minimum_amount CHECK ((minimum_amount >  
(0)::numeric)),  
    CONSTRAINT positive_minimum_term CHECK ((minimum_term > 0))  
);
```

```
ALTER TABLE bank.credit_type OWNER TO postgres;
```

```
--
```

```

-- TOC entry 221 (class 1259 OID 16399)
-- Name: currency; Type: TABLE; Schema: bank; Owner: postgres
--

CREATE TABLE bank.currency (
    id integer NOT NULL,
    name character varying(20) NOT NULL
);

ALTER TABLE bank.currency OWNER TO postgres;

--
-- TOC entry 226 (class 1259 OID 16461)
-- Name: deposit_contract; Type: TABLE; Schema: bank; Owner: postgres
--

CREATE TABLE bank.deposit_contract (
    id integer NOT NULL,
    date_start date NOT NULL,
    date_end date NOT NULL,
    amount numeric(12,2) NOT NULL,
    currency_id integer NOT NULL,
    deposit_type_id integer NOT NULL,
    employee_id integer NOT NULL,
    client_id integer NOT NULL,
    CONSTRAINT positive_deposit_amount CHECK ((amount > (0)::numeric)),
    CONSTRAINT valid_deposit_dates CHECK ((date_end > date_start))
);

ALTER TABLE bank.deposit_contract OWNER TO postgres;

--
-- TOC entry 230 (class 1259 OID 16563)
-- Name: deposit_interest; Type: TABLE; Schema: bank; Owner: postgres
--

CREATE TABLE bank.deposit_interest (
    id integer NOT NULL,
    deposit_contract_id integer NOT NULL,
    month integer NOT NULL,
    accrued_amount numeric(12,2) NOT NULL,
    accrued_interest numeric(12,2) NOT NULL,
    CONSTRAINT deposit_interest_amount_nonnegative CHECK ((accrued_amount >=
(0)::numeric)),

```

```

        CONSTRAINT deposit_interest_interest_nonnegative CHECK ((accrued_interest
>= (0)::numeric)),
        CONSTRAINT deposit_interest_month CHECK ((month >= 1))
);

```

```

ALTER TABLE bank.deposit_interest OWNER TO postgres;

```

```

--
-- TOC entry 224 (class 1259 OID 16435)
-- Name: deposit_type; Type: TABLE; Schema: bank; Owner: postgres
--

```

```

CREATE TABLE bank.deposit_type (
    id integer NOT NULL,
    name character varying(50) NOT NULL,
    description character varying(255),
    minimum_term integer NOT NULL,
    minimum_amount numeric(12,2) NOT NULL,
    interest_rate numeric(5,2) NOT NULL,
    CONSTRAINT positive_interest_rate CHECK ((interest_rate >= (0)::numeric)),
    CONSTRAINT positive_minimum_amount CHECK ((minimum_amount >
(0)::numeric)),
    CONSTRAINT positive_minimum_term CHECK ((minimum_term > 0))
);

```

```

ALTER TABLE bank.deposit_type OWNER TO postgres;

```

```

--
-- TOC entry 223 (class 1259 OID 16419)
-- Name: employee; Type: TABLE; Schema: bank; Owner: postgres
--

```

```

CREATE TABLE bank.employee (
    id integer NOT NULL,
    name character varying(100) NOT NULL,
    phone_number character varying(20),
    address character varying(150),
    id_position integer NOT NULL,
    passport character varying(11) NOT NULL,
    CONSTRAINT employee_phone_check CHECK (phone_number ~ '^\\+?[0-9]{11}$'),
    CONSTRAINT employee_passport_check CHECK (passport ~ '^[0-9]{4}\\s?[0-
9]{6}$')
);

```



```
ALTER TABLE bank.employee OWNER TO postgres;
```

```
--  
-- TOC entry 229 (class 1259 OID 16550)  
-- Name: payment_fact; Type: TABLE; Schema: bank; Owner: postgres  
--
```

```
CREATE TABLE bank.payment_fact (  
    id integer NOT NULL,  
    is_overdue boolean NOT NULL,  
    payment_date date,  
    payment_id integer NOT NULL  
);
```

```
ALTER TABLE bank.payment_fact OWNER TO postgres;
```

```
--  
-- TOC entry 228 (class 1259 OID 16531)  
-- Name: payment_schedule; Type: TABLE; Schema: bank; Owner: postgres  
--
```

```
CREATE TABLE bank.payment_schedule (  
    id integer NOT NULL,  
    contract_id integer NOT NULL,  
    month_number integer NOT NULL,  
    payment_date date NOT NULL,  
    principal_amount numeric(12,2) NOT NULL,  
    interest_amount numeric(12,2) NOT NULL,  
    CONSTRAINT positive_interest_amount CHECK ((interest_amount >=  
(0)::numeric)),  
    CONSTRAINT "positive_month_number " CHECK ((month_number >= 1)),  
    CONSTRAINT "positive_principal_amount " CHECK ((principal_amount >=  
(0)::numeric))  
);
```

```
ALTER TABLE bank.payment_schedule OWNER TO postgres;
```

```
--  
-- TOC entry 220 (class 1259 OID 16390)  
-- Name: position; Type: TABLE; Schema: bank; Owner: postgres  
--
```

```

CREATE TABLE bank."position" (
    id integer NOT NULL,
    "name" character varying(50) NOT NULL,
    salary numeric(10,2) NOT NULL,
    CONSTRAINT salary_positive CHECK ((salary > (0)::numeric))
);

ALTER TABLE bank."position" OWNER TO postgres;

--
-- TOC entry 5103 (class 0 OID 16408)
-- Dependencies: 222
-- Data for Name: client; Type: TABLE DATA; Schema: bank; Owner: postgres
--

COPY bank.client (id, name, phone_number, email, passport, address) FROM
stdin;
1  Иванов Иван Иванович      +79990000001      ivanov1@mail.ru 4010 123456 Санкт-
Петербург, Невский пр., 1
2  Петров Петр Петрович      +79990000002      petrov2@mail.ru 4011 123457 Санкт-
Петербург, Литейный пр., 15
3  Сидорова Анна Викторовна  +79990000003      sidorova3@mail.ru 4012
123458 Санкт-Петербург, Московский пр., 24
4  Кузнецов Алексей Сергеевич +79990000004      kuznetsov4@mail.ru 4013
123459 Санкт-Петербург, ул. Марата, 7
5  Смирнова Мария Олеговна  +79990000005      smirnova5@mail.ru 4014 123460
Санкт-Петербург, ул. Савушкина, 18
\.

--
-- TOC entry 5108 (class 0 OID 16496)
-- Dependencies: 227
-- Data for Name: credit_contract; Type: TABLE DATA; Schema: bank; Owner:
postgres
--

COPY bank.credit_contract (id, date_start, date_end, client_id, employee_id,
currency_id, amount, credit_type_id) FROM stdin;
1  2026-04-01  2026-10-01  2  2  1  100000.00  1
2  2026-02-01  2027-02-01  4  2  1  450000.00  2
3  2026-01-10  2031-01-10  5  2  1  2500000.00  3
\.

```

```
--
-- TOC entry 5106 (class 0 OID 16448)
-- Dependencies: 225
-- Data for Name: credit_type; Type: TABLE DATA; Schema: bank; Owner: postgres
--
```

```
COPY bank.credit_type (id, name, description, minimum_amount, minimum_term,
interest_rate) FROM stdin;
```

```
1 Потребительский Кредит на личные нужды 30000.00 6 19.90
2 Автокредит Кредит на покупку автомобиля 200000.00 12 14.50
3 Ипотека Кредит на недвижимость 1000000.00 60 11.20
\.
```

```
--
-- TOC entry 5102 (class 0 OID 16399)
-- Dependencies: 221
-- Data for Name: currency; Type: TABLE DATA; Schema: bank; Owner: postgres
--
```

```
COPY bank.currency (id, name) FROM stdin;
```

```
1 RUB
2 USD
3 EUR
\.
```

```
--
-- TOC entry 5107 (class 0 OID 16461)
-- Dependencies: 226
-- Data for Name: deposit_contract; Type: TABLE DATA; Schema: bank; Owner:
postgres
--
```

```
COPY bank.deposit_contract (id, date_start, date_end, amount, currency_id,
deposit_type_id, employee_id, client_id) FROM stdin;
```

```
1 2026-04-01 2026-10-01 120000.00 1 1 1 1
2 2026-03-15 2027-03-15 300000.00 1 2 1 3
3 2026-04-05 2026-07-05 15000.00 1 3 3 5
\.
```

```
--
-- TOC entry 5111 (class 0 OID 16563)
-- Dependencies: 230
```

```

-- Data for Name: deposit_interest; Type: TABLE DATA; Schema: bank; Owner:
postgres
--

COPY bank.deposit_interest (id, deposit_contract_id, month, accrued_amount,
accrued_interest) FROM stdin;
1 1 4 850.00 8.50
2 1 5 850.00 8.50
3 2 3 2500.00 10.00
4 3 4 90.00 7.20
\.

--
-- TOC entry 5105 (class 0 OID 16435)
-- Dependencies: 224
-- Data for Name: deposit_type; Type: TABLE DATA; Schema: bank; Owner:
postgres
--

COPY bank.deposit_type (id, name, description, minimum_term, minimum_amount,
interest_rate) FROM stdin;
1 Накопительный Пополняемый вклад 6 10000.00 8.50
2 Срочный Без пополнения 12 50000.00 10.00
3 Пенсионный Льготный вклад 3 5000.00 7.20
\.

--
-- TOC entry 5104 (class 0 OID 16419)
-- Dependencies: 223
-- Data for Name: employee; Type: TABLE DATA; Schema: bank; Owner: postgres
--

COPY bank.employee (id, name, phone_number, address, id_position, passport)
FROM stdin;
1 Орлова Наталья Игоревна +79000000001 Санкт-Петербург, ул. Типанова, 10
1 5001 111111
2 Федоров Максим Андреевич +79000000002 Санкт-Петербург, ул.
Бассейная, 21 2 5002 111112
3 Егорова Елена Павловна +79000000003 Санкт-Петербург, ул. Варшавская,
33 3 5003 111113
\.

--

```

```

-- TOC entry 5110 (class 0 OID 16550)
-- Dependencies: 229
-- Data for Name: payment_fact; Type: TABLE DATA; Schema: bank; Owner:
postgres
--

COPY bank.payment_fact (id, is_overdue, payment_date, payment_id) FROM stdin;
1  f   2026-05-01  1
2  t   \N  2
3  f   2026-03-01  3
4  f   2026-04-02  4
\.

--

-- TOC entry 5109 (class 0 OID 16531)
-- Dependencies: 228
-- Data for Name: payment_schedule; Type: TABLE DATA; Schema: bank; Owner:
postgres
--

COPY bank.payment_schedule (id, contract_id, month_number, payment_date,
principal_amount, interest_amount) FROM stdin;
1  1  1   2026-05-01  15000.00   2500.00
2  1  2   2026-06-01  15000.00   2200.00
3  2  1   2026-03-01  30000.00   5000.00
4  2  2   2026-04-01  30000.00   4700.00
5  3  1   2026-02-10  20000.00   15000.00
6  3  2   2026-03-10  20000.00   14800.00
\.

--

-- TOC entry 5101 (class 0 OID 16390)
-- Dependencies: 220
-- Data for Name: position; Type: TABLE DATA; Schema: bank; Owner: postgres
--

COPY bank."position" (id, "name ", salary) FROM stdin;
1  Менеджер      70000.00
2  Кредитный специалист  85000.00
3  Консультант  60000.00
\.

--

```

```
-- TOC entry 4921 (class 2606 OID 16418)
-- Name: client client_passport_unique; Type: CONSTRAINT; Schema: bank; Owner:
postgres
--

ALTER TABLE ONLY bank.client
    ADD CONSTRAINT client_passport_unique UNIQUE (passport);

--

-- TOC entry 4923 (class 2606 OID 16416)
-- Name: client client_pkey; Type: CONSTRAINT; Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.client
    ADD CONSTRAINT client_pkey PRIMARY KEY (id);

--

-- TOC entry 4935 (class 2606 OID 16510)
-- Name: credit_contract credit_contract_pkey; Type: CONSTRAINT; Schema: bank;
Owner: postgres
--

ALTER TABLE ONLY bank.credit_contract
    ADD CONSTRAINT credit_contract_pkey PRIMARY KEY (id);

--

-- TOC entry 4931 (class 2606 OID 16460)
-- Name: credit_type credit_type_pkey; Type: CONSTRAINT; Schema: bank; Owner:
postgres
--

ALTER TABLE ONLY bank.credit_type
    ADD CONSTRAINT credit_type_pkey PRIMARY KEY (id);

--

-- TOC entry 4917 (class 2606 OID 16407)
-- Name: currency currency_name_unique; Type: CONSTRAINT; Schema: bank; Owner:
postgres
--

ALTER TABLE ONLY bank.currency
```

```
ADD CONSTRAINT currency_name_unique UNIQUE (name);

--
-- TOC entry 4919 (class 2606 OID 16405)
-- Name: currency currency_pkey; Type: CONSTRAINT; Schema: bank; Owner:
postgres
--

ALTER TABLE ONLY bank.currency
    ADD CONSTRAINT currency_pkey PRIMARY KEY (id);

--
-- TOC entry 4933 (class 2606 OID 16475)
-- Name: deposit_contract deposit_contract_pkey; Type: CONSTRAINT; Schema:
bank; Owner: postgres
--

ALTER TABLE ONLY bank.deposit_contract
    ADD CONSTRAINT deposit_contract_pkey PRIMARY KEY (id);

--
-- TOC entry 4941 (class 2606 OID 16575)
-- Name: deposit_interest deposit_interest_pkey; Type: CONSTRAINT; Schema:
bank; Owner: postgres
--

ALTER TABLE ONLY bank.deposit_interest
    ADD CONSTRAINT deposit_interest_pkey PRIMARY KEY (id);

--
-- TOC entry 4929 (class 2606 OID 16447)
-- Name: deposit_type deposit_type_pkey; Type: CONSTRAINT; Schema: bank;
Owner: postgres
--

ALTER TABLE ONLY bank.deposit_type
    ADD CONSTRAINT deposit_type_pkey PRIMARY KEY (id);

--
-- TOC entry 4925 (class 2606 OID 16429)
```

```
-- Name: employee employee_passport_unique; Type: CONSTRAINT; Schema: bank;
Owner: postgres
--

ALTER TABLE ONLY bank.employee
    ADD CONSTRAINT employee_passport_unique UNIQUE (passport);

--

-- TOC entry 4927 (class 2606 OID 16427)
-- Name: employee employee_pkey; Type: CONSTRAINT; Schema: bank; Owner:
postgres
--

ALTER TABLE ONLY bank.employee
    ADD CONSTRAINT employee_pkey PRIMARY KEY (id);

--

-- TOC entry 4939 (class 2606 OID 16557)
-- Name: payment_fact payment_fact_pkey; Type: CONSTRAINT; Schema: bank;
Owner: postgres
--

ALTER TABLE ONLY bank.payment_fact
    ADD CONSTRAINT payment_fact_pkey PRIMARY KEY (id);

--

-- TOC entry 4937 (class 2606 OID 16544)
-- Name: payment_schedule payment_schedule_pkey; Type: CONSTRAINT; Schema:
bank; Owner: postgres
--

ALTER TABLE ONLY bank.payment_schedule
    ADD CONSTRAINT payment_schedule_pkey PRIMARY KEY (id);

--

-- TOC entry 4915 (class 2606 OID 16398)
-- Name: position position_pkey; Type: CONSTRAINT; Schema: bank; Owner:
postgres
--

ALTER TABLE ONLY bank."position"
```



```

ADD CONSTRAINT position_pkey PRIMARY KEY (id);

--
-- TOC entry 4947 (class 2606 OID 16511)
-- Name: credit_contract credit_contract_currency_fk; Type: FK CONSTRAINT;
Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.credit_contract
    ADD CONSTRAINT credit_contract_currency_fk FOREIGN KEY (currency_id)
REFERENCES bank.currency(id);

--
-- TOC entry 4948 (class 2606 OID 16526)
-- Name: credit_contract credit_contract_employee_fk; Type: FK CONSTRAINT;
Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.credit_contract
    ADD CONSTRAINT credit_contract_employee_fk FOREIGN KEY (employee_id)
REFERENCES bank.employee(id);

--
-- TOC entry 4949 (class 2606 OID 16516)
-- Name: credit_contract credit_contract_type_fk; Type: FK CONSTRAINT; Schema:
bank; Owner: postgres
--

ALTER TABLE ONLY bank.credit_contract
    ADD CONSTRAINT credit_contract_type_fk FOREIGN KEY (credit_type_id)
REFERENCES bank.credit_type(id);

--
-- TOC entry 4950 (class 2606 OID 16521)
-- Name: credit_contract credt_contract_client_fk; Type: FK CONSTRAINT;
Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.credit_contract
    ADD CONSTRAINT credt_contract_client_fk FOREIGN KEY (client_id) REFERENCES
bank.client(id);

```

```
--
-- TOC entry 4943 (class 2606 OID 16486)
-- Name: deposit_contract deposit_contract_client_fk; Type: FK CONSTRAINT;
Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.deposit_contract
    ADD CONSTRAINT deposit_contract_client_fk FOREIGN KEY (client_id)
REFERENCES bank.client(id);

--
-- TOC entry 4944 (class 2606 OID 16476)
-- Name: deposit_contract deposit_contract_currency_fk; Type: FK CONSTRAINT;
Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.deposit_contract
    ADD CONSTRAINT deposit_contract_currency_fk FOREIGN KEY (currency_id)
REFERENCES bank.currency(id);

--
-- TOC entry 4945 (class 2606 OID 16491)
-- Name: deposit_contract deposit_contract_employee_fk; Type: FK CONSTRAINT;
Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.deposit_contract
    ADD CONSTRAINT deposit_contract_employee_fk FOREIGN KEY (employee_id)
REFERENCES bank.employee(id);

--
-- TOC entry 4946 (class 2606 OID 16481)
-- Name: deposit_contract deposit_contract_type_fk; Type: FK CONSTRAINT;
Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.deposit_contract
    ADD CONSTRAINT deposit_contract_type_fk FOREIGN KEY (deposit_type_id)
REFERENCES bank.deposit_type(id);
```

```
--
-- TOC entry 4953 (class 2606 OID 16576)
-- Name: deposit_interest deposit_interest_contract_fk; Type: FK CONSTRAINT;
Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.deposit_interest
    ADD CONSTRAINT deposit_interest_contract_fk FOREIGN KEY
(deposit_contract_id) REFERENCES bank.deposit_contract(id);

--
-- TOC entry 4942 (class 2606 OID 16430)
-- Name: employee employee_position_fk; Type: FK CONSTRAINT; Schema: bank;
Owner: postgres
--

ALTER TABLE ONLY bank.employee
    ADD CONSTRAINT employee_position_fk FOREIGN KEY (id_position) REFERENCES
bank."position"(id);

--
-- TOC entry 4952 (class 2606 OID 16558)
-- Name: payment_fact payment_fact_schedule_fk; Type: FK CONSTRAINT; Schema:
bank; Owner: postgres
--

ALTER TABLE ONLY bank.payment_fact
    ADD CONSTRAINT payment_fact_schedule_fk FOREIGN KEY (payment_id)
REFERENCES bank.payment_schedule(id);

--
-- TOC entry 4951 (class 2606 OID 16545)
-- Name: payment_schedule payment_schedule_contract_fk; Type: FK CONSTRAINT;
Schema: bank; Owner: postgres
--

ALTER TABLE ONLY bank.payment_schedule
    ADD CONSTRAINT payment_schedule_contract_fk FOREIGN KEY (contract_id)
REFERENCES bank.credit_contract(id);
```

```
-- Completed on 2026-04-05 18:35:52

--
-- PostgreSQL database dump complete
--

\unrestrict zHfzVYNHwb6rxDaU6YUMbY1q8VzHSjoQFRuHLBvpvEsoPPyVqJhaAQKvY8jHipL
```

Выводы:

Выводы: в ходе выполнения лабораторной работы была реализована база данных PostgreSQL для предметной области «Банк». Были созданы таблицы в соответствии с разработанной моделью данных, настроены ограничения целостности (первичные и внешние ключи, ограничения уникальности и проверки допустимых значений), выполнено заполнение таблиц рабочими данными. Также были освоены операции резервного копирования и восстановления базы данных средствами pgAdmin 4. Получены практические навыки проектирования и реализации реляционных баз данных.